Section 1.4 Separable Equations and Applications

What is a separable differential equation?

> The differential equation $\frac{dy}{dx} = f(x, y)$ is called seperable if there are two functions $g(x), k(y)$ where g only depends on x and k only depends on y such that $f(x, y) = g(x)k(y)$.

Example Which of the following differential equations are separable? Identify $g(x) and\ k(y)$ if they are separable.

$\frac{dy}{dx} = xy - 3y$

$\frac{dy}{dx} = xy^2 - 3y$

$\frac{dP}{dx} = kP(M - P)$

$\frac{dv}{dt} = 32 - kv$

$\frac{dy}{dx} = x^2 - y^2$

$\frac{dy}{dx} = \frac{x}{y}$

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Solving Separable Differential Equations

First remember the chain rule! $\frac{d}{dx}H\big(y(x)\big) = H'\big(y\big)\frac{dy}{dx}$ if we let $h(y) = H'(y)$ we can write the chain rule as:

$$\frac{d}{dx}H\big(y(x)\big) = h\big(y\big)\frac{dy}{dx}$$

Let $\frac{dy}{dx} = g(x)k(y)$ how can we re-write this so that we can simply integrate bothsides with respect x and end up with an equation we can solve for y with? Let $k\big(y\big) = 1/h(y)$…

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Example: Find the general solution to the differential equation: $\frac{dy}{dx} = ky$

Example Find the particular solution to the differential equation $\frac{dy}{dx} = 2xy^2 + 3x^2y^2,\ y(1) = 1$

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Implicit Solutions. There will be some D.E.'s for which we cannot directly solve for y.

Let $H(y) = G(x) + c$ be the result of using seperation of variables on the differential equation $\frac{dy}{dx} = f(x, y)$ and suppose we can or don't solve for y *interms of* x we can them represent the solution implicitly with $H(y) - G(x) - C = 0$

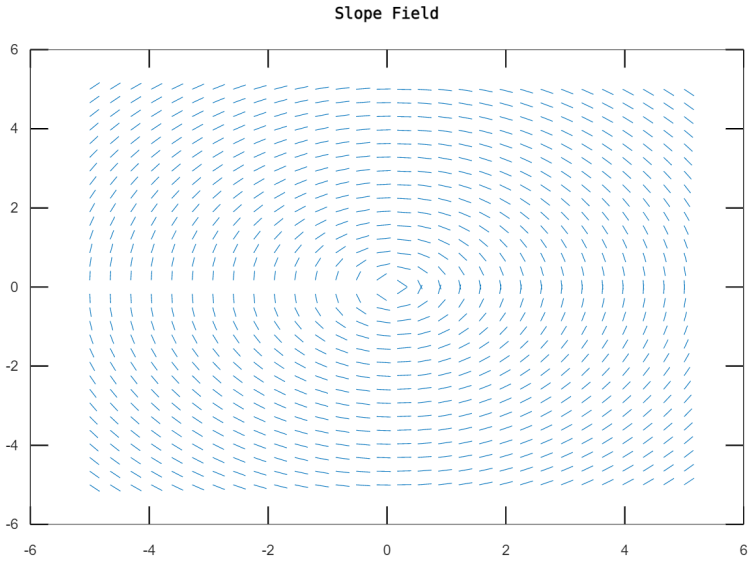
If we name $K(x, y) = H(y) - G(x) - C = 0$ we call K the implicit solution of the differential equation.

Example: Solve the differential equation $\frac{dy}{dx} = \frac{-x}{y}$

Let's also look at the slope field.

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In the last example we wrote down the implicit solution to $\frac{dy}{dx} = -\frac{x}{y}$ as $y^2 + x^2 - C = 0$ the equation of a circle. Suppose we are given the initial data $y(0) = 2$ what does this tell us about the particular solution?



We must be careful though! We note that $\frac{dy}{dx} = -\frac{x}{y}$ is not defined at $y = 0$. This means that a solution to our D.E. cannot cross $y = 0$. So the picture of a full circle is not quite the correct one. Because of our derivative we need to restrict our solution to never cross the x-axis. So suppose we are given two initial data $y(0) = 2$ and $y(0) = -2$. How could we represent the solution to our differential equation?

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Some Applications

The natural growth equation. $\frac{dy}{dt} = ky$ We have seen this pop up a lot so far and that is because it can be used to model any phenomina for whith the relevant quantity changes in proportion to that quantity. For example:

Population Growth: Suppose that b and d are the birth and death rates. Then the change in population P over the time Δt would be

$\Delta P = b\Delta \text{t}P - d\Delta \text{t}P = (b-d)\Delta \text{t}P$

Then $\frac{\Delta P}{\Delta t} = (b-d)P$

And thus $\lim_{\Delta_t \to 0} \frac{\Delta P}{\Delta t} = \frac{dp}{dt} = (b-d)P$

Or just $\frac{dp}{dt} = kp \;\; k > 0$

Radioactive decay
Suppose element A decays into Element B at a rate r *over a time* Δt and that B never turns into A. Then the change in the number of of atoms, N, of element A over time Δt
Would be

$\Delta N = 0\Delta \text{t}N - r\Delta \text{t}N = (-r)\Delta \text{t}N$

Then $\frac{\Delta N}{\Delta t} = (-r)N$

And thus $\lim_{\Delta_t \to 0} \frac{\Delta P}{\Delta t} = \frac{dp}{dt} = -rN$

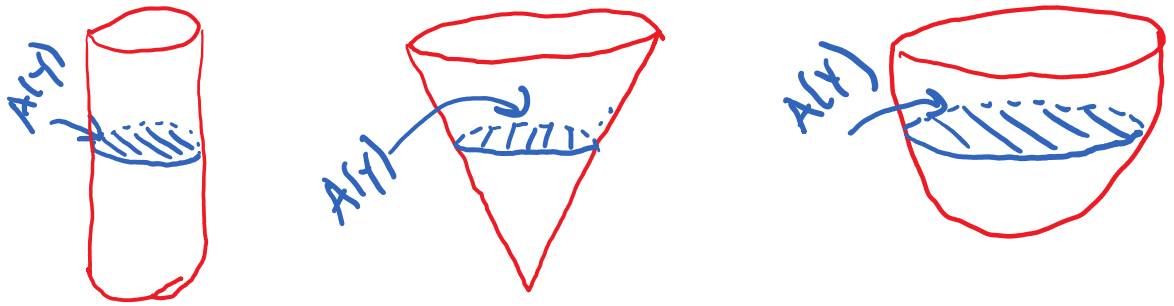
Or just $\frac{dp}{dt} = -kp \;\; k > 0$

This equation shows up for compound interest, drug elimination in the body, viral spread and more! Weather something is growing or decaying just depends on the sign of the constant on the right hand side.

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Torricellie's law states that the change in the volume of a liquid like water in a tank that is being drained through a hole in the bottom is given by: $\frac{dV}{dt} = -a\sqrt{2gy}$,where $y(t)$ is depth of the liquid at time t.

Note that here we have two functions of time V and y and so we need to rewrite our DE interms of just one of them in order to be able to solve it. Luckily integral calculus has us covered. Recall that we can find the volume of a solid by integrating along its center axis as a function of is crossectional area.



$V = \int A(y)dy$ Remember that an integral is a sum and here we are adding up all the crossectional areas to get a volume! What happens when we take the derivative of this equation with respect to y

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Let's suppose the tank was shaped like a parabola $y = x^2$. What would our differential equation look like?

$$A(y)\frac{dy}{dt} = -a\sqrt{2gy}$$

What is the general solution?

How could we get an equation for the volume from this?

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Finally, Lets solve the logistic equation $\frac{dy}{dt} = ky(M - y)$.